

The determinant as the source of the cubic term: normal-form reduction in a matrix gradient flow

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Self-contained manuscript in the language of dynamical systems; it does not require any external framework. The Spanish version is in paper_es.md. Reproducible verification scripts, one per numerical statement (§7), are in code/.

Abstract

We consider the gradient flow $\dot{\Gamma} = -\nabla P(\Gamma; \mu)$ on the real 4×4 matrices, with potential $P(\Gamma) = \|\Gamma\|^2 + \mu \det \Gamma + \beta \|\Gamma\|^4$ (plus an optional sextic regularizer), and its damped second-order counterpart $\ddot{\Gamma} + \gamma \dot{\Gamma} + \nabla P = 0$. Near a degeneration of the Hessian —a simple soft mode— center-manifold reduction yields the local normal forms of bifurcation theory. The cubic coefficient of the reduced flow comes from the determinant through its cofactor matrix, the only anisotropic nonlinearity of the field; it is the sole source of the cubic when the soft mode is orthogonal to Γ_* . We classify the accessible organizing centers: fold and pitchfork in codimension 1, the cusp in the gradient sector, and Bogdanov–Takens in codimension 2, whose existence is topologically obstructed in the gradient limit and requires lifting the system to the second-order (inertial) flow. The homoclinic orbit closes the Bogdanov–Takens portrait, and the non-gradient part of the field, which carries the rotation, numerically sustains a Shilnikov-type chaotic regime. The analytical results are accompanied by simulations that verify the critical scalings: the Kramers law, a pseudo-arclength continuation through the fold, the 3/2 law of the cusp, and the Lyapunov exponent of the chaotic regime.

Keywords: matrix gradient flow, center-manifold reduction, normal forms, cofactor matrix, cusp, Bogdanov–Takens, Shilnikov chaos.

1. Introduction

Bifurcation theory organizes the qualitative changes of a dynamical system around a catalogue of normal forms: the fold $\dot{\xi} = \lambda - \xi^2$, the pitchfork $\dot{\xi} = \lambda\xi - \xi^3$, the cusp, Hopf, Bogdanov–Takens (Guckenheimer–Holmes 1983; Kuznetsov 2004). We study the geometric role of the determinant in selecting the normal form for matrix-valued gradient flows when the Hessian degenerates along a soft mode.

Under hypotheses (H1)–(H4), the determinant generates the cubic term of the local reduction. Its gradient is the cofactor matrix $C(\Gamma) = \text{cof}(\Gamma)$, multilinear in the entries and anisotropic, whereas the norm terms $\|\Gamma\|^{2k}$ are isotropic. Upon projection onto the center manifold of a soft mode, this anisotropy survives as the cubic coefficient of the normal form, and persists even when the norm terms vanish, i.e. when the soft mode is orthogonal to Γ_* .

We prove a codimension-1 reduction theorem with explicit coefficients (§3–§4); we classify the two codimension-2 organizing centers, cusp and Bogdanov–Takens, and show that the latter requires the second-order dynamics (§5); we analyze the loss of gradient structure and its relation to chaos (§6); and we support each result with reproducible numerical verification (§7).

2. The system

Let $V = M_4(\mathbb{R}) \cong \mathbb{R}^{16}$ with the Frobenius inner product $\langle X, Y \rangle = \text{tr}(X^\top Y)$. We take

$$P(\Gamma; \mu, J) = \|\Gamma\|^2 + \mu \det \Gamma + \beta \|\Gamma\|^4 + b_6 \|\Gamma\|^6 - \langle J, \Gamma \rangle, \quad \beta \geq 0, b_6 \geq 0,$$

where $\mu \in \mathbb{R}$ is the control parameter and $J \in M_4(\mathbb{R})$ an external field. The sextic b_6 only bounds the branches globally and plays no role in the local statements. The field-free potential ($J = 0$) depends only on $\|\Gamma\|^2$ and $\det \Gamma$, both invariant under the bilateral orthogonal action $\Gamma \mapsto U\Gamma V^\top$ with $U, V \in O(4)$ (the same action as the singular-value decomposition, which reappears in §4). This large isotropy makes its degeneracies occur in families rather than in isolation. A generic field J breaks it and places the equilibrium in a configuration where the soft mode of the bifurcation is simple (hypothesis H2): the linear term $\langle J, \Gamma \rangle$ intersects the $O(4) \times O(4)$ orbits transversally (Thom transversality), lifting the degeneracy imposed by the continuous symmetry and leaving a single zero eigenvalue. Since J enters linearly, it shifts the equilibrium but does not alter the Hessian. The gradient flow is

$$\dot{\Gamma} = -\nabla P, \quad \nabla P = 2\Gamma + \mu C(\Gamma) + (4\beta \|\Gamma\|^2 + 6b_6 \|\Gamma\|^4)\Gamma,$$

with $C(\Gamma) = \text{cof}(\Gamma) = \partial \det \Gamma / \partial \Gamma$. The Hessian $H(\Gamma) = D^2 P$ is symmetric since the field is a gradient. We also consider the damped second-order version

$$\ddot{\Gamma} + \gamma \dot{\Gamma} + \nabla P = 0, \quad \gamma \in \mathbb{R},$$

whose overdamped limit $\gamma \rightarrow \infty$ recovers the gradient flow. The system is deterministic; only to probe the basins of attraction and the escape rates (§7) do we resort, auxilarily, to the stochastic Langevin extension $\dot{\Gamma} = -\nabla P + \sqrt{2D} \eta(t)$, with η matrix white noise and $D > 0$.

Remark 2.1 (scope). Parts 1 and 2 of Theorem 1 are the center-manifold / Lyapunov–Schmidt reduction for any real-analytic P under (H1)–(H2), and do not depend on the specific form of P ; only the identification of the cubic coefficient (§3, item 3) and of the effective quartic use the particular potential. The construction requires a positive-definite inner product—the Frobenius one—which bounds P from below and fixes the gradient structure, and is independent of any later metric reinterpretation of the entries of Γ . A Monte Carlo sampling over the coefficients (β, b_6) and the field J confirms that the simple soft mode and the contribution of the determinant persist, indicating structural stability of the codimension-1 reduction.

3. Main result (codimension 1)

We assume at the point (Γ_*, μ_*) the following hypotheses:

- (H1) $\nabla P(\Gamma_*, \mu_*) = 0$;
- (H2) $H_* = H(\Gamma_*)$ has a simple zero eigenvalue, with unit eigenvector V , and the rest of the spectrum bounded away from zero;
- (H3) transversality: $\tau := \langle V, C(\Gamma_*) \rangle \neq 0$, so the control μ moves the soft mode;
- (H4) nondegeneracy: $a_3 := D^3 P(\Gamma_*)[V, V, V] \neq 0$; or else a $\mathbb{Z}_2$ involution forces $a_3 = 0$ and then $a_4^{\text{eff}} \neq 0$.

Theorem 1 (reduction $\Gamma \rightarrow \xi$). Under (H1)–(H2) there exists, in a neighborhood of (Γ_*, μ_*) , a unique, smooth one-dimensional center manifold

$\Gamma = \Gamma_* + \xi V + h(\xi, \mu)$ with $h \in V^\perp$ and $h = O(\xi^2, \xi\tilde{\mu})$, on which the flow is gradient, $\dot{\xi} = -\partial_\xi \Phi(\xi, \mu)$. If moreover (H3)–(H4) hold:

1. if $a_3 := D^3P(\Gamma_*)[V, V, V] \neq 0$, the reduced form is the fold
 $\dot{\xi} = \alpha - \frac{1}{2}a_3\xi^2$, with $\alpha = -\tau(\mu - \mu_*)$ and $\tau = \langle V, C(\Gamma_*) \rangle$;
2. if a $\mathbb{Z}_2$ involution forces $a_3 = 0$, the reduced form is the pitchfork
 $\dot{\xi} = -a'_2\xi - \frac{1}{6}a_4^{\text{eff}}\xi^3$;
3. $a_3 = \mu_* D^3\det(\Gamma_*)[V, V, V] + 24\beta\langle \Gamma_*, V \rangle$; within the space of invariants of the potential (1), the first term is the sole source of the cubic when $V \perp \Gamma_*$ (a degree-three invariant such as $\text{tr } \Gamma^3$, absent in (1), would also contribute);
4. $a_4^{\text{eff}} = D^4P[V^{\otimes 4}] - 3\langle D^3P[V, V], (H_*|_{V^\perp})^{-1}D^3P[V, V] \rangle$, whose sign depends on the spectrum of $H_*|_{V^\perp}$ (positive-definite on the stable branch, indefinite at a saddle).

In the symmetric sector, a_4^{eff} changes sign at $\mu = 16\beta$, the line separating the single-equilibrium regime from the three-equilibrium one (Figure 1).

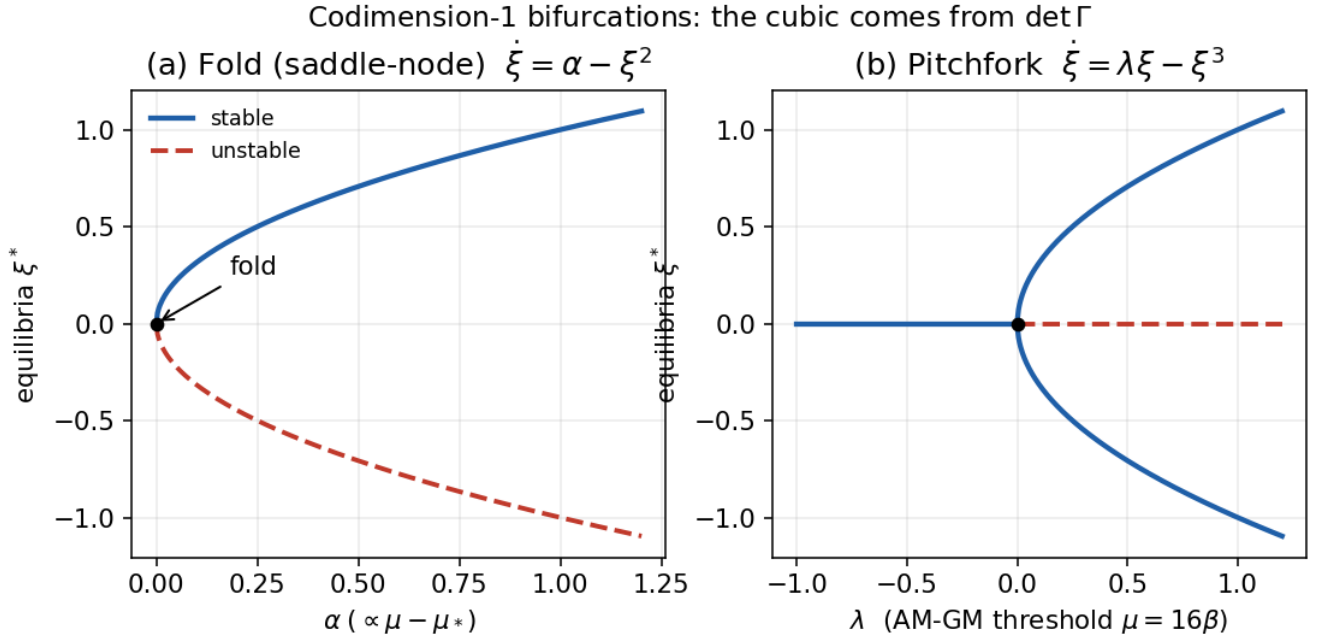


Figure 1: Figure 1. Codimension-1 bifurcations: fold $\dot{\xi} = \alpha - \xi^2$ and pitchfork $\dot{\xi} = \lambda\xi - \xi^3$. Stable branches in blue, unstable in dashed red. The cubic term comes from $\det \Gamma$.

4. Proof (Lyapunov–Schmidt)

The reduction is the Lyapunov–Schmidt one for gradient fields (Carr 1981; Golubitsky–Schaeffer 1985). Write $\Gamma = \Gamma_* + \xi V + W$ with $W \in V^\perp$ and $\mu = \mu_* + \nu$, and let Q be the orthogonal projection onto $V^\perp = \text{ran } H_*$. The equation $\nabla P = 0$ splits into its components in $V^\perp$ and in V .

The component in $V^\perp$, $Q\nabla P = 0$, is solved by the implicit function theorem — $H_*|_{V^\perp}$ is invertible by (H2)— and yields the slaving manifold $W = h(\xi, \mu)$, with $h(0, \mu_*) = 0$, $\partial_\xi h(0, \mu_*) = 0$ and $\partial_\xi^2 h(0) = -(H_*|_{V^\perp})^{-1} Q D^3P(\Gamma_*)[V, V]$ upon differentiating twice. Substituting h into the remaining component gives the bifurcation equation

$g(\xi, \mu) = \langle V, \nabla P \rangle$, which —since the field is a gradient— equals $\partial_\xi \Phi$, $\Phi = P|_{\text{center mfld}}$; the reduced flow $\dot{\xi} = -g$ inherits the gradient structure. Writing $a_k = \partial_\xi^k \Phi(0, \mu_*)$:

- $\partial_\nu a_1|_0 = \langle V, \partial_\mu \nabla P \rangle = \langle V, C(\Gamma_*) \rangle = \tau$
(the second term $\langle V, H_* \partial_\mu h \rangle$ vanishes because $H_* \partial_\mu h \perp V$);
- $a_2(\mu_*) = \langle V, H_* V \rangle = 0$ (soft mode);
- $a_3 = \langle V, D^3 P[V, V] \rangle = D^3 P(\Gamma_*)[V, V, V]$; the slaving term $\langle V, H_* \partial_\xi^2 h \rangle$ vanishes since it is $\perp V$, so the cubic receives no correction.

To identify a_3 one decomposes $D^3 P$ term by term. The $\|\Gamma\|^2$ term is quadratic and does not contribute. For the quartic norm, with $\|V\| = 1$, $D^3(\|\Gamma\|^4)[V, V, V] = 24 \langle \Gamma_*, V \rangle$. For the determinant, which is degree 4 in 4×4 , $D^3 \det$ is linear and $D^3(\mu \det \Gamma)[V, V, V] = \mu_* D^3 \det(\Gamma_*)[V, V, V]$. Summing,

$$a_3 = \mu_* D^3 \det(\Gamma_*)[V, V, V] + 24\beta \langle \Gamma_*, V \rangle,$$

which is item 3. The first term is generically nonzero even when $\langle \Gamma_*, V \rangle = 0$: there the norm is silent and, within the invariants of (1), the determinant is the sole source of the cubic. This has a direct geometric reading in terms of the singular values of Γ . The determinant is their product, $\det \Gamma = \prod_i \sigma_i$ (up to sign), while the norm is their sum of squares, $\|\Gamma\|^2 = \sum_i \sigma_i^2$. The product couples the four singular values irreducibly and supplies the anisotropic part of the cubic; the sum is isotropic and contributes only through $\langle \Gamma_*, V \rangle$. The effective quartic is obtained by differentiating g once more and substituting $\partial_\xi^2 h$, giving item 4; the sign of the correction is fixed by the definiteness of $H_*|_{V^\perp}$. The fold/pitchfork classification then follows from Sotomayor's conditions (Sotomayor 1973; Kuznetsov 2004): with $a_1(\mu_*) = a_2(\mu_*) = 0$ and $a_3 \neq 0$ one gets the fold $\dot{\xi} = \alpha - \frac{1}{2}a_3\xi^2$; if a $\mathbb{Z}_2$ symmetry annihilates a_3 (and $\langle \Gamma_*, V \rangle$), the leading nontrivial term is the quartic and one gets the pitchfork. $\square$

5. Codimension-2 organizing centers

Theorem 2 (cusp, A_3). The point $a_2 = a_3 = 0$ with $a_4^{\text{eff}} \neq 0$ is a cusp, the universal unfolding of the pitchfork (Thom 1975; Golubitsky–Schaeffer 1985). In this system a_3 changes sign, because the determinant and norm contributions compete, so the point exists; adding $a_2 = 0$ fixes it. A two-parameter family (μ, s) —control and field amplitude— unfolds it versally, with $\partial(a_1, a_2)/\partial(\mu, s)$ nonsingular. The three-equilibrium window has width $\propto (-a_2)^{3/2}$ (Figure 2).

Theorem 3 (Bogdanov–Takens). In the gradient flow this bifurcation is obstructed: the Jacobian $-H_*$ is symmetric, hence diagonalizable, and a double zero eigenvalue would have geometric multiplicity 2 rather than a Jordan block. Releasing the Jordan block requires a non-gradient field; within the dissipative phenomenology we study, this is achieved by lifting the system to the second-order equation. There the linearization $\begin{pmatrix} 0 & I \\ -H_* & -\gamma I \end{pmatrix}$ has, on the soft mode, eigenvalues $\{0, -\gamma\}$, which at $\gamma = 0$ collapse to a double zero with a Jordan block. The reduction is $\ddot{\xi} + \gamma(\xi)\dot{\xi} + (a_1 + c\xi^2) = 0$ with $c = \frac{1}{2}a_3$; with state-dependent damping $\gamma(\xi) = \gamma_0 + \gamma_1\xi$, the saddle-node (the fold), Hopf ($\gamma_0 = -\gamma_1\xi_*$) and homoclinic curves emanate from the BT point (Bogdanov 1975; Takens 1974).

The saddle-node branch of Bogdanov–Takens coincides with the fold of Theorem 1, so both codimension-2 centers are degeneracies of the same soft mode. The BT point marks the boundary between the overdamped

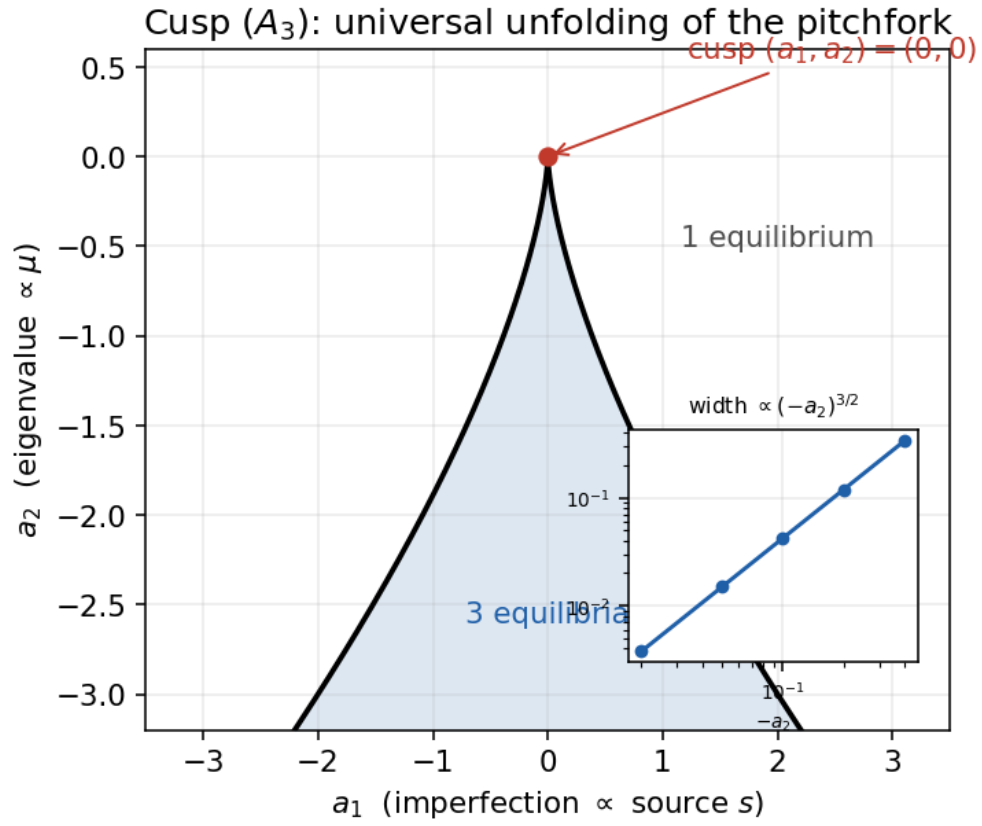


Figure 2: Figure 2. The cusp A_3 in the unfolding plane (a_1, a_2) . Inside the semicubical wedge $4a_2^3 + 27a_1^2 \leq 0$ there are three equilibria; outside, one. The inset shows the $3/2$ law.

regime —gradient, with fold and cusp— and the oscillatory one (Figure 3).

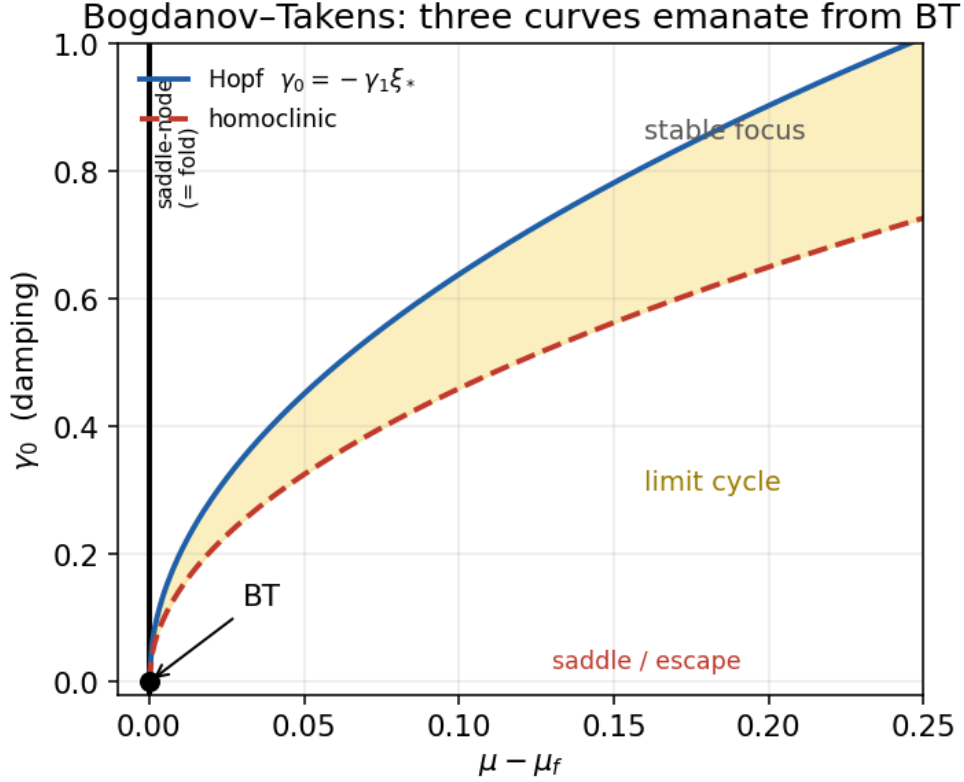


Figure 3: Figure 3. Bogdanov-Takens unfolding in $(\mu - \mu_f, \gamma_0)$: the saddle-node (the fold), Hopf and homoclinic curves emanate from BT; the limit cycle lives between the Hopf and homoclinic curves.

6. Non-gradient extension and chaos phenomenology

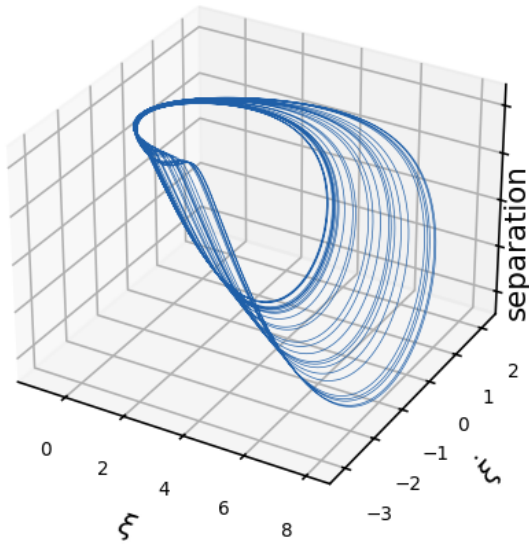
Proposition 4 (energy obstruction). The second-order equation with uniform $\gamma \geq 0$ admits $E = \frac{1}{2}\|\dot{\Gamma}\|^2 + P$ as a Lyapunov function, since $\dot{E} = -\gamma\|\dot{\Gamma}\|^2 \leq 0$; every trajectory relaxes to an equilibrium, so the dissipative gradient system admits no sustained chaos.

Chaos therefore cannot emerge from the mere dissipation of the coupled degrees of freedom: it requires an energy input ($\gamma_{\text{eff}} \leq 0$ in some region) paired with the asymmetry of the reactive sector Γ_a , which breaks the symmetry of the Hessian and introduces the rotation ($\text{Im } \lambda \neq 0$). That antisymmetric, non-gradient part also supplies the third dimension that Poincaré-Bendixson requires. Coupling it with active damping, the cubic nonlinearity induced by the determinant produces a jerk equation with a saddle-focus equilibrium; integrating it, one observes an attractor with positive Lyapunov exponent ($\lambda \approx 0.055$) and a limit cycle that ends in a homoclinic with logarithmically divergent period (Figure 4). The reduction of the second-order flow in $16 + 16$ dimensions to this jerk equation is not proved here; the evidence is numerical, and we state it as a conjecture.

Conjecture 1. In the inertial extension coupled with the antisymmetric part of the field, the cubic nonlinearity induced by the determinant suffices to produce a homoclinic connection to a saddle-focus equilibrium satisfying the Shilnikov condition ($|\lambda_{\text{real}}| > |\text{Re } \lambda_{\text{compl}}|$; Shilnikov 1965), and with it a chaotic attractor. The analytical unfolding of this global bifurcation remains open.

Reactive-sector chaos: quadratic jerk, $\lambda_{\max} \approx 0.055$

(a) Chaotic attractor (saddle-focus)



(b) Sensitive dependence

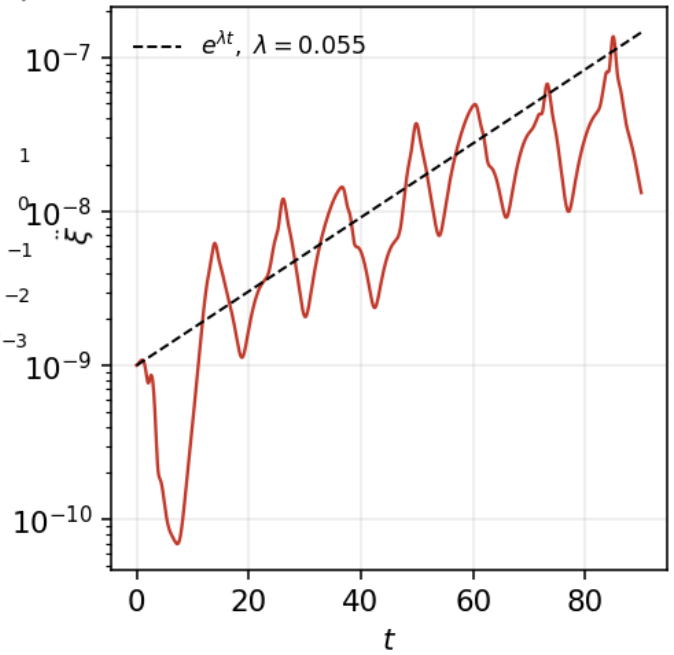


Figure 4: Figure 4. (a) Shilnikov-type chaotic attractor (saddle-focus) of the quadratic jerk equation. (b) Sensitive dependence: the separation of two trajectories grows as $e^{\lambda t}$ with $\lambda \approx 0.055$.

7. Numerical verification

Each statement is accompanied by a self-contained Python (NumPy/SciPy) script that verifies it reproducibly. Degeneration points are located by solving $\nabla P = 0$ and the soft-mode condition $\lambda_{\min}(H) = 0$ simultaneously; normal-form coefficients are extracted by polynomial fitting of P restricted to the relevant directions; Lyapunov exponents are computed by the Benettin method (Benettin et al. 1980); escape rates are compared with the Kramers law (Hänggi et al. 1990); and equilibrium branches are followed by pseudo-arclength continuation (Doedel 1981). The table summarizes object, result and script.

#	object	result	script
1	fold/pitchfork as invariant objects; $\text{Var} \sim 1/k$; Monte Carlo 10^3	genuine critical pitchfork $k_2 = 0$; $k \cdot \text{Var} \approx \text{const}$; sorting of 10^3 matrices at 100%	code/pieza1_bifurcaciones_rigor.py
2	exact reduction (invariant ray)	$\nabla P \parallel \Gamma$; P_{red} pitchfork; threshold $\mu = 16\beta$	code/pieza1_reduccion_normal_forms.py
3	generic center manifold	$B = 2 \sum g_i^2 / \omega_i - b$; full flow = reduced (not the naive one)	code/pieza1_centro_manifold_generico.py
4	Kramers + continuation	$\ln \langle \tau \rangle = 0.904 \Delta U / D + 1.78$, $R^2 = 0.989$; fold traced to $\mu \approx 2.04$	code/pieza1_kramers_continuacion.py
5	Theorem 1 in 16-dim	simple zero; $\tau = -12.84$; $a_3 = 5.09$ (det -11.67); real saddle-node	code/pieza1_teorema_4x4.py
6	Theorem 2 (cusp)	$a_2, a_3 \sim 10^{-11}$; versal Jacobian $= 1.30$; $3/2$ law: $\propto (-a_2)^{1.503}$, $R^2 = 1.0$	code/pieza1_cuspide_codim2.py
7	Theorem 3 (BT)	Jordan block; Hopf curve; limit cycle	code/pieza1_bogdanov_takens.py
8	homoclinic + chaos	$T = 0.738(-\ln \Delta) + 2.08$, $R^2 = 1.0$; Lyapunov $\lambda \approx 0.055$ (Sprott)	code/pieza1_homoclinica_caos.py
9	energy obstruction + chaos from the EOM	$\gamma \geq 0 \Rightarrow E$ Lyapunov $\Rightarrow$ relaxes; active damping $\Rightarrow$ limit cycle (2 modes)	code/pieza1_caos_EOM_2modos.py
10	robustness / structural stability	varying (a, β, b_6, J) : 40/40 keep simple soft mode, fold and determinant contribution	code/pieza1_robustez_teorema.py

8. Discussion

The central content is that, when the dominant nonlinearity of a matrix gradient flow is the determinant, the geometry of the matrix —through its cofactor— determines the type of bifurcation of the soft mode. The sequence fold $\rightarrow$ cusp $\rightarrow$ Bogdanov–Takens $\rightarrow$ chaos is traversed by degenerating a single mode and adding, at each step, a structure: the determinant supplies the cubic that distinguishes fold from pitchfork; the damping γ supplies the second Bogdanov–Takens parameter upon passing to second order; and the non-gradient part supplies the rotation that opens the door to chaos. The determinant enters as the invariant that generates the cubic, not as a stability indicator: linear stability is fixed by the spectrum of the symmetric Hessian, while $\det \Gamma$ encodes orientation and rank.

Three lines remain open. First, the rigorous reduction of the second-order flow in $16 + 16$ dimensions to the reactive jerk equation; the non-gradient model in §6 is illustrative, not an exact

reduction. Second, the full unfolding of the chaotic scenario —cascades, the Shilnikov–Hopf connection— in the non-gradient sector. Third, the extension to $M_n(\mathbb{R})$ with $n > 4$, where $\det$ has degree n and supplies higher-order nonlinearities.

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